

HealthConnect final analytics and decision support

Week 8 final integration package | AnalystLab Africa Experience Lab

Wilson Moses | Data Analytics perspective with cross-track integration to the existing Data Science work

Project: Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Executive summary

The supplied HealthConnect file contains **5,000 appointment records** for **1,696 distinct anonymized patients**. Of 4,737 completed appointments, 2,423 were recorded as no-shows, giving a no-show rate of **51.2%**. Cancellations (263, 5.3% of all records) are reported separately. Booking lead time shows the clearest descriptive gradient: 26.6% for bookings 0-3 days ahead and 63.9% for bookings 31 or more days ahead.

A supportive confirmation or reminder test for long-lead appointments is a reasonable next question. The dataset does not prove an intervention will work. It is synthetic and observational, patient IDs repeat, and the reviewed materials did not include a Week 7 Data Analytics report. This report does not claim Week 7 tests or human cross-track approval.

1. Scope and continuity

This output turns the existing HealthConnect appointment data into a final decision-support view. It addresses how HealthConnect could use data and AI to reduce missed appointments and improve patient support. It does not start a new project or replace the source data. The original CSV and dictionary are copied unchanged into the package.

Track context

Available Week 4-6 records describe Wilson Moses work in the Data Science track, including a no-history logistic regression candidate with mean temporal validation AUC 0.6481. The attached Week 8 assignment is specifically for Data Analytics. This package treats the analytics findings as a cross-track contribution. No Week 7 Analytics report or test result was available, so that transition remains an open evidence item.

2. Data and method

The analysis uses HealthConnect_Appointment_Data.csv and its data dictionary. It contains 5,000 appointment records from 01 Jan 2025 to 30 Jun 2026. Appointment ID is unique. Patient IDs repeat, so appointments are the unit of analysis and are not independent people. Rates are descriptive and no significance tests or causal claims are made.

KPI definitions

KPI	Definition	Result
Appointment records	All rows in source	5,000
Completed appointments	Attended + No-Show	4,737
Attendance rate	Attended / completed	48.8%
No-show rate	No-Show / completed	51.2%
Cancellation rate	Cancelled / all records	5.3%

The no-show denominator excludes cancellations. The cancellation rate uses all appointment records.

3. Data validation

Core consistency checks passed: appointment IDs are unique, no exact duplicate rows were found, booking lead days match the difference between the two dates, weekday labels match appointment dates, previous no-show counts do not exceed prior appointments, and all outcome labels match the stated categories. The data dictionary permits limited missing distance values.

Check	Observed	Status
Exact duplicate rows	0	PASS
Appointment ID duplicates	0	PASS
Lead-day/date mismatches	0	PASS
Weekday/date mismatches	0	PASS
Missing distance	90 (1.8%)	DOCUMENTED
Missing waiting time	60 (1.2%)	DOCUMENTED
Repeated patients	1,696 distinct patients across 5,000 appointments	DOCUMENTED

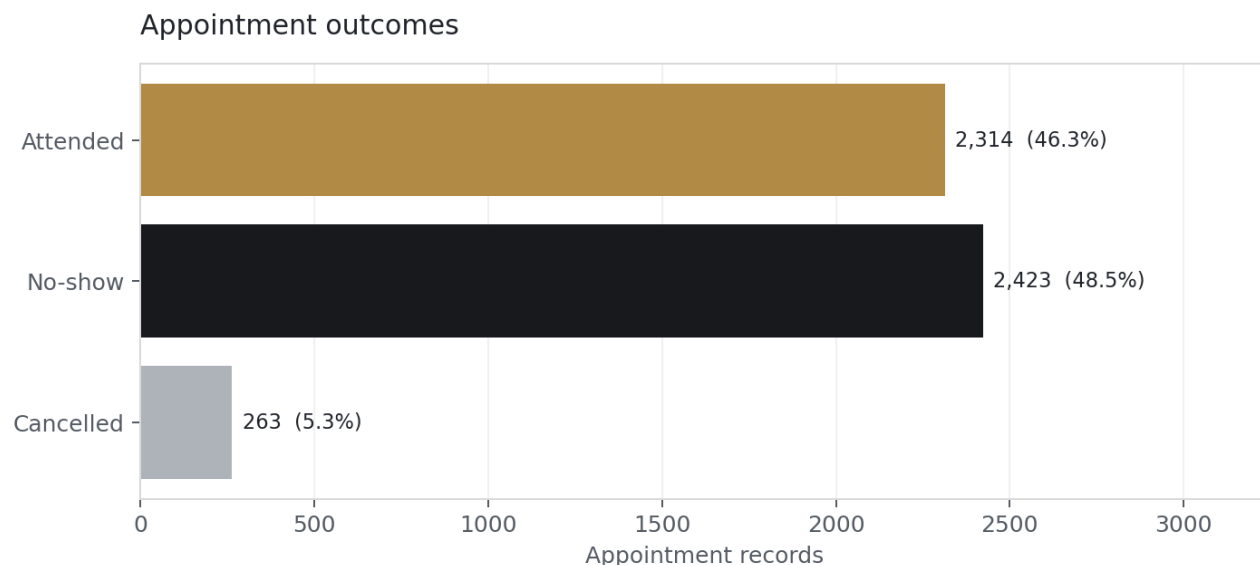
Waiting time is excluded from booking-time prediction recommendations because its availability point is unclear and it could be recorded after the appointment process begins.

Field interpretation

Field	Use in this analysis	Caution
Appointment outcome	No-show rate uses Attended + No-Show. Cancellations have a separate denominator.	Do not combine cancellations with completed visits.
Booking lead time	Grouped as 0-3, 4-7, 8-14, 15-30 and 31+ days. Source matches date differences.	Rates describe the supplied groups.
Reminder status and channel	Compared descriptively for completed appointments.	Assignment, timing, delivery and response are not controlled.
Distance to clinic	Retained as a source field.	90 values (1.8%) are missing.
Waiting time	Validated for missingness only.	60 values (1.2%) are missing; timing is unclear.
Patient ID	Used to count distinct patients and repeated records.	The same patient can have multiple appointments.

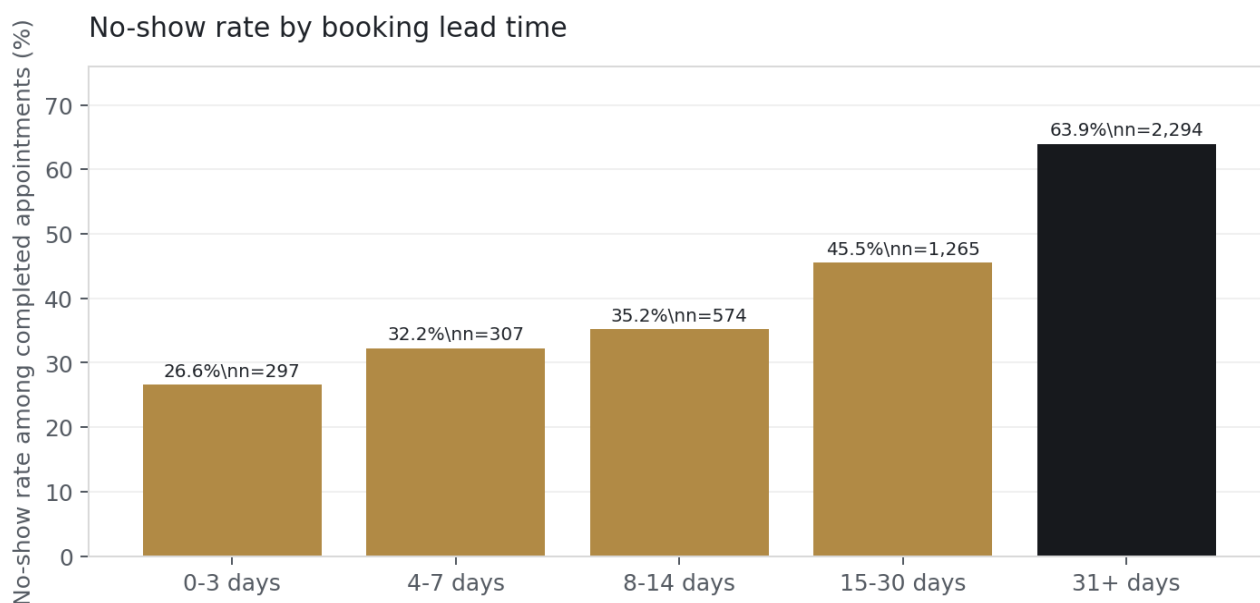
4. What the appointment data shows

Outcome mix



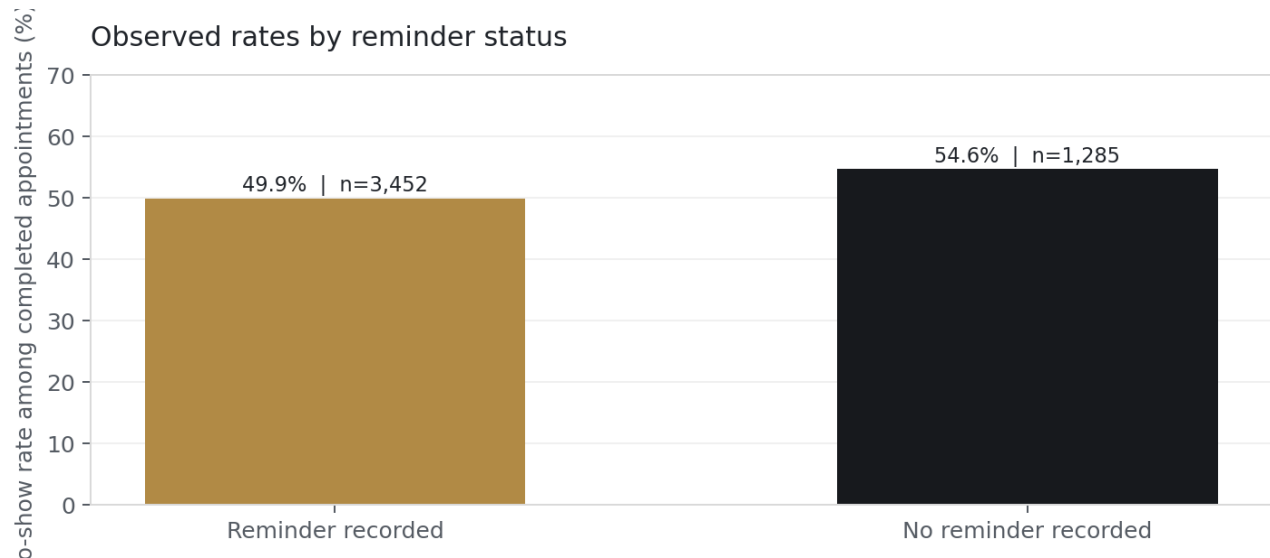
2,314 appointments were attended, 2,423 were no-shows and 263 were cancelled. This describes the fictional project file, not real clinic performance.

Booking lead time



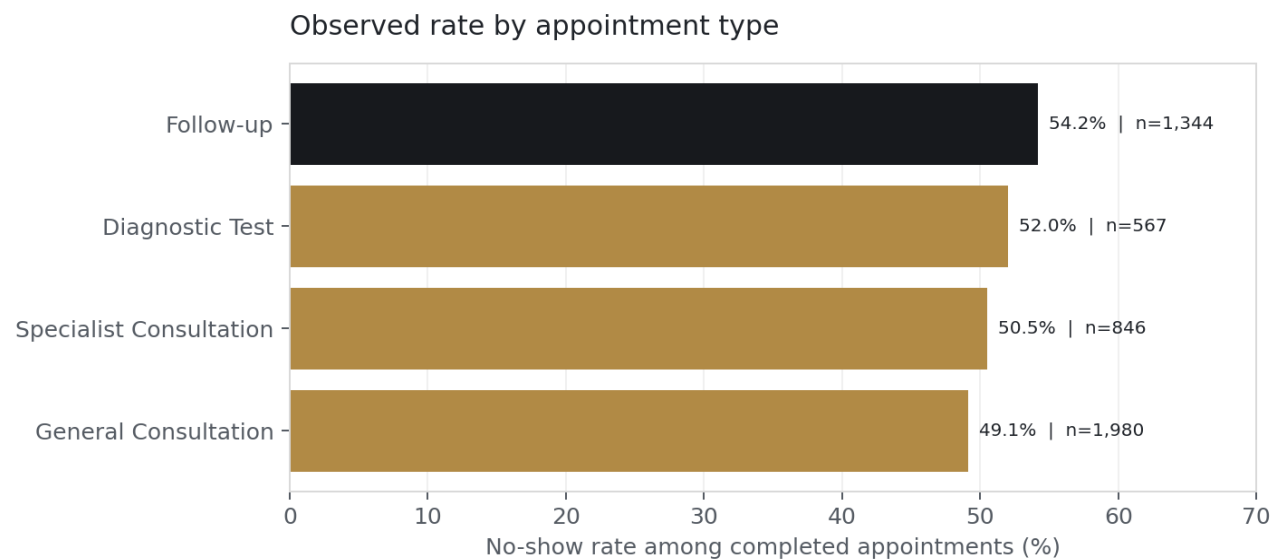
The rate rises across lead-time groups, from 26.6% for bookings 0-3 days ahead to 63.9% for 31+ days. The longest-lead group includes 2,294 completed appointments and 1,467 no-shows. These account for 60.5% of all no-shows. This supports a prospective review of confirmation timing.

Reminder status



Reminder coverage was 72.7% across all records. Among completed appointments, the recorded no-show rate was 49.9% when a reminder was recorded and 54.6% when none was recorded. The raw difference is 4.8 percentage points. Assignment, delivery, timing and patient response are uncontrolled, so this is an association.

Appointment type



Rates range from 49.1% for General Consultation to 54.2% for Follow-up. Use appointment type to describe service mix, not as evidence of cause.

5. Business insights and recommendations

Finding	Evidence	Recommended use
Long lead time is a priority to test	31+ days: 63.9%; 1,467 no-shows	Test confirmation closer to the visit. Measure attendance and early slot recovery.
Reminder records show a lower rate	49.9% with a reminder recorded vs 54.6% without	Review timing and delivery. Test before attributing impact.
Cancellations need a separate workflow	263 records (5.3% of all records)	Track cancellation lead time and whether released slots are refilled.
Appointment-type differences are modest	Follow-up 54.2%; General Consultation 49.1%	Use as operational context, not a causal rule.

Pilot measures

Define the eligible appointment population and contact rules before testing. Compare a new confirmation or reminder approach with current practice. Track completed attendance, cancellations, successful contact, refilled slots, staff time, complaints and opt-outs. Where feasible, randomize by appointment and account for repeat patients in analysis.

Avoid using small demographic differences for punitive decisions. The data are synthetic, observations repeat by patient, and subgroup rates do not establish a reason for missing an appointment.

6. Cross-track integration

The analytics package provides explicit rate definitions and lead-time segments to the Data Science workstream. The available Week 6 summary records a no-history logistic regression candidate with mean temporal validation AUC 0.6481. That model result remains distinct from the descriptive percentages here. Analytics can help the model team interpret lead time and define a monitoring slice. The model team must confirm feature availability at prediction time and test on fresh data.

Direction	Output and use	Evidence status
Analytics to Data Science	KPI definitions, lead-time and reminder segments to support interpretation	Prepared for handoff; partner review not recorded
Data Science to Analytics	Week 6 model candidate and AUC context	Prior summary available; Week 7 update not located
Analytics to Project Management	Executive findings and pilot measures	Included in package; receipt not recorded
Week 7 Analytics to Week 8	Test result, issue, refinement and ready component	Week 7 Analytics record not supplied

This report records a prepared handoff. It does not claim another track or HC-POD reviewed or approved the output.

7. Readiness and limits

Ready for final presentation as a descriptive decision-support component. Not validated for operational deployment or clinical decisions. No live data feed, patient outreach rule or measured attendance improvement is claimed.

Open before a real pilot

Attach the Week 7 Analytics report if available. Obtain partner-track review of the rate definitions. Confirm real appointment and cancellation workflows, permitted contact channels, reminder delivery logs and staffing capacity. Establish fresh time-based evaluation before using predictive scores to allocate staff attention.

The companion script and notebook regenerate the KPI tables and figures. The original raw source was not edited.